

I am a passionate educator and researcher with a PhD in Computer Engineering and a diverse international academic background. My expertise spans Natural Language Processing, Generative AI, and Machine Learning, with a growing focus on Cybersecurity applications in intelligent systems. I have taught a wide range of undergraduate and graduate courses across institutions in Canada, Europe, and the Middle East, and have mentored numerous graduate students. My research contributions include impactful publications, successful grant proposals, and interdisciplinary collaborations. I am committed to fostering inclusive learning environments, advancing AI and cybersecurity research, and building academic-industry partnerships that drive innovation and societal impact.

EDUCATION

- **PhD in Computer Engineering** - (WES-approved) 2012–2016
Eastern Mediterranean University, Cyprus
Dissertation: *Chemical Named Entity Recognition through Ensemble Learning and Particle Swarm Optimization*
Advisor: Prof. Ekrem Varoglu
- **Visiting PhD student** Fall 2014
Xidian University, Xian, China.
- **MSc in Software Engineering** (WES-approved) 2006–2008, IAU, Tehran, Iran
- **BSc in Software Engineering** (WES-approved) 2003–2006, University of Tabriz, Iran

RESEARCH & TEACHING EXPERIENCE

- **Adjunct Professor** September 2025 - Present
Carleton University, Ottawa, b
Course: Deep Learning
- **AI Consultant & Lecturer** Summer 2025
Algonquin College, Ottawa, Canada
- **Course thought:** Natural Language Processing
- **Postdoctoral Fellow** 2023–2025
Carleton University, Ottawa, Canada
- Research focus: Generative AI for Accommodation Recommendation
- Collaborated on NFRF project (PI: Prof. Majid Komeili)
- Mentored 2 master and 2 PhD students
- **Assistant Professor** 2022 - 2023
Department of Computer Engineering, Istanbul Gelisim University, Turkey
- **Undergraduate courses:** Object Oriented programming, Introduction to machine learning, Programming 1
- **Graduate courses:** Data Mining
- **Research project:** Participating in a research project related to 3d printers which uses metal bullets instead of filament.
- **Visiting Teaching Professor** 2021 - 2023 (Hybrid)
Department of Computer Engineering, Cyprus Science University, Cyprus
- **Undergraduate courses:** Programming II, Data structure, Artificial Intelligence
- **Graduate courses:** Machine Learning, Advanced Neural Networks
- **Lecturer** 2020 - 2021
Department of Computer Engineering, Islamic Azad University, Iran
- **Undergraduate courses:** Data Mining, Object Oriented Programming, Undergraduate projects
- **Graduate courses:** Natural Language Processing, Research Methods
- **Postdoctoral Associate** 2018–2020
KULeuven, Leuven, Belgium
- Developed a Hierarchical Classification Model for ICD Coding
- Developed a novel Model for Causal Relation Extraction in Biomedical Text
- Collaborated on ACCUMULATE project(PI: Prof. Marie-Francine Moens) - Mentored 1 master student
- **Postdoctoral Researcher** 2017–2018
University of Zagreb, Zagreb, Croatia
- Developed a novel model for Word Sense Induction
- Mentored 1 master student
- Collaborated on SenseHive project (PI: prof. Jan Snajder)
- **PhD Study** 2012–2016
In my PhD research, I tackled the Named Entity Recognition (NER) challenge in chemical texts using an ensemble learning approach. This involved training around 500 recognizers to form a diverse selection pool. I also developed a heuristic Particle Swarm Optimization (PSO)-based algorithm for recognizer selection. Additionally, I introduced two innovative methodologies: a chemical domain-specific tokenizer (ChemTok) and a sentence-based undersampling technique aimed at addressing the class imbalance issue prevalent in this field.

SELECTED PUBLICATION

Peer-Reviewed Journal Articles

- Zarei, M. R., Akkasi, A., & Komeili, M. (2025). Dual Adapter Tuning of Vision–Language Models Using Large Language Models. *International Journal of Computational Intelligence Systems*, 18(1), 109. Springer.
- Khan, A., Choubineh, A., Shaaban, M. A., Akkasi, A., & Komeili, M. (2025). TactileNet: Bridging the Accessibility Gap with AI-Generated Tactile Graphics for Individuals with Vision Impairment. *arXiv preprint arXiv:2504.04722*.
- Akkasi, A. (2024). Job description parsing with explainable transformer based ensemble models to extract the technical and non-technical skills. *Natural Language Processing Journal*, 9, 100102. <https://doi.org/10.1016/j.nlp.2024.100102>
- Shaaban, M. A., Akkasi, A., Khan, A., Komeili, M., & Yaqub, M. (2024). Fine-tuned large language models for symptom recognition from Spanish clinical text. *arXiv preprint arXiv:2401.15780*.
- Akkasi, A., & Snajder, J. (2021). Word sense induction using leader-follower clustering of automatically generated lexical substitutes. *Expert Systems with Applications*, 181, 115162.
- Akkasi, A., & Moens, M.-F. (2021). Causal relationship extraction from biomedical text using deep neural models: A comprehensive survey. *Journal of Biomedical Informatics*, 119, 103820.
- Moons, E., Khanna, A., Akkasi, A., & Moens, M.-F. (2020). A comparison of deep learning methods for ICD coding of clinical records. *Applied Sciences*, 10(15), 5262.
- Akkasi, A., & Varoglu, E. (2019). Improvement of chemical named entity recognition through sentence-based random under-sampling and classifier combination. *Journal of AI and Data Mining*, 7(2), 311–319.
- Harounabadi, M., Mitschele-Thiel, A., & Akkasi, A. (2018). LTE-D2D for connected cars: a survey on radio resource management schemes. *Iran Journal of Computer Science*, 1(3), 187–197.
- Akkasi, A. (2018). Sentence-based undersampling for named entity recognition using genetic algorithm. *Iran Journal of Computer Science*, 1(3), 165–174.
- Akkasi, A., Varoğlu, E., & Dimililer, N. (2018). Balanced undersampling: a novel sentence-based undersampling method to improve recognition of named entities in chemical and biomedical text. *Applied Intelligence*, 48(8), 1965–1978.
- Akkasi, A., & Varoğlu, E. (2016). Improving biochemical named entity recognition using PSO classifier selection and Bayesian combination methods. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 14(6), 1327–1338.
- Akkasi, A. (2016). Chemical Named Entity Recognition using Undersampling and Classifier Ensembles. Eastern Mediterranean University (EMU).
- Zarezade, S., Akkasi, A., Maher, A., & Namdar, Y. (2015). Recognition of Farsi Letter using Hidden Markov Model. *International Journal of Computer Applications*, 125(1).
- Asadi, M. H., Akkasi, A., Zargarpour, E., & Mohammadi, Z. (2015). Diagnosis of Mathematical Symbols using Hidden Markov Model. *International Journal of Computer Applications*, 125(1).
- Sattari, S., Akkasi, A., Lari, R. A., & Khodaparasti, M. (2015). Cryptography in social networks using wavelet transform, fractals and chaotic functions. *International Research Journal of Applied and Basic Sciences*, 1627–1635.
- Pishdar, M. A., & Akkasi, A. (2015). Task scheduling and idle-time balancing in homogeneous multiprocessors: A comparison between GA and SA. *International Journal of Computer Applications*, 123(13).
- Akkasi, A., & Zeinaly, M. A. (2015). Visualizing Operating Systems and Markov Models. *International Journal of Computer Applications*, 119(24).
- Akkasi, A., Khaleghi, A., Jafarabad, M., Karimi, H., Demideh, M. B., & Alamuti, R. N. (2014). The Proposal of Giving Two Receipts for Voters to Increase the Security of Electronic Voting. *Computer Science & Information Technology (CS & IT)*.

Conference and Workshop Proceedings

- Kevadiya, R., Akkasi, A., Fraser, K. C., Vukovic, B., Gunnell, J., Tanguay, S., & Komeili, M. (2025). Generalized Follow-Up WHO-DAS 2.0 Assessment Through Language Models and Adaptive Clustering Ensemble in Higher Education. In *Proceedings of the 18th ACM International Conference on Pervasive Technologies Related to Assistive Environments (PETRA '25)*, 349–358. <https://doi.org/10.1145/3733155.3734900>
- Akkasi, A., Khan, A., Shaaban, M. A., Komeili, M., & Yaqub, M. (2024). iML at SemEval-2024 Task 2: Safe Biomedical Natural Language Interference for Clinical Trials with LLM Based Ensemble Inferencing. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, 170–174.
- Akkasi, A., Fraser, K. C., & Komeili, M. (2023). Reference-free summarization evaluation with large language models. In *Proceedings of the 4th Workshop on Evaluation and Comparison of NLP Systems*, 193–201.
- Akkasi, A. (2022). Multi perspective scientific document summarization with graph attention networks (GATS). In *Proceedings of the Third Workshop on Scholarly Document Processing*, 268–272.
- Gluhak, M., Di Buono, M. P., Akkasi, A., Šnajder, J., et al. (2018). TakeLab at SemEval-2018 Task 7: Combining Sparse and Dense Features for Relation Classification in Scientific Texts. In *Proceedings of the 12th Workshop on Semantic Evaluation (SemEval-2018)*.
- Akkasi, A., Shams, F., et al. (2008). Presenting A Method for Benchmarking Application in the Enterprise Architecture Planning Process Based on Federal Enterprise Architecture Framework. In *2008 3rd International Conference on Information*

and Communication Technologies: From Theory to Applications, 1–6. IEEE.

Preprints / Under Review

- Dastjerdi, Mohammad Mahdi Heydari, et al. "End-to-End Semantically Aware Tactile Generation." *Intelligent Systems With Applications*, 2025.

I also have 15 papers presented at internal conferences in Iran, which are accessible via this [link](#). One can enter my name and family name to see the list of publications.

PROJECTS

• Academic Accommodation Recommendation Engine

Technologies: Python, PyTorch, LangChain, Azure AI, Prompt Engineering, LM Studio

- Designed and built a **Retrieval-Augmented Generation (RAG)** system to recommend personalized academic accommodations for students based on their profiles and needs.
- Integrated Azure AI Services for secure, scalable deployment.
- **Improved recommendation** relevance by 12% compared to manual rule-based methods through advanced prompt engineering and model fine-tuning. Moreover, it provides interpretability by explaining the rationale behind its decisions.
- Under **deployment by Carleton University** for students with disability and accessibility institute. It will be used by Fall 2025.

• Banking Customer Service Chatbot with Agentic RAG

Technologies: RAG, LlamaIndex, Open-Source LLMs, LangGraph, LangChain, Vector Databases

- Built and deployed an RAG-based chatbot to handle general banking queries, reducing call center workload by 23%.
- Implemented context-aware retrieval to ensure accurate and compliant answers for financial services.
- **Deployed by Eghtesad Novin Bank, Iran.**

• Deep Learning-Based ICD Coding of Clinical Records

Technologies: Python, TensorFlow, CNN, RNN, Hierarchical Classification

- Evaluated and benchmarked deep learning models for automated multi-label classification of clinical discharge summaries into ICD-9 and ICD-10 codes.
- Incorporated the hierarchical structure of ICD codes into model design, boosting classification accuracy.
- Applied few-shot learning techniques using category description vectors to improve performance.
- The model's performance improved significantly, outperforming traditional machine learning and deep learning models by at least 7%.
- **Deployed** as part of Hospital Management System (HMS) at **University Hospital, Leuven, Belgium.**

• Automated Skill Network Generator for Job Descriptions

Technologies: AWS SageMaker, Python, TensorFlow, LIME, spaCy, Multi-Task Learning, Data Scraping & Annotation

- Created an **end-to-end AWS SageMaker pipeline** to parse job descriptions and automatically build skill relationship graphs.
- Applied ensemble learning and transformer-based NER models to improve skill extraction accuracy.
- Created and annotated a custom dataset of over 22,000 job postings across 22 IT roles from LinkedIn, Glassdoor, and Indeed.
- Incorporated LIME explainability to provide interpretable predictions, improving trust in model outputs.
- Achieved F1-scores of 72% (technical skills) and 67% (non-technical skills), significantly outperforming baseline models.
- Designed to use to create a Skills Cloud, enhance job matching, career guidance, and workforce planning in recruitment processes.
- Deployed as part of the Skills Cloud system by Begreator startup, Belgium.

• Prompt-Optimized Text Summarization Evaluator

Technologies: HuggingFace Transformers, Prompt Engineering, LangChain, LlamaIndex

- Designed a summary quality evaluation framework using Chain-of-Thought and Tree-of-Thoughts prompting strategies.
- Achieved more consistent evaluation scores compared to standard metrics like ROUGE and BLEU by incorporating semantic similarity checks.
- Demonstrated the possibility of evaluating generated summaries without ground truth.
- Proposed prompting strategy won the **Best Solution Award** at the 4th ACM Workshop on Evaluation and Comparison of NLP Systems.

• Multimodal Visual Question Answering System

Technologies: BLIP-2, PEFT, LORA, Vision-Language Models, Multimodal AI, LangChain, Pydantic

- Created a parameter-efficient fine-tuned VQA pipeline combining vision-language models with LLM reasoning.
- Developed a dual adapter architecture combining CLIP-Adapter (prior-dependent) and Task-Residual (prior-independent)

modules.

- Integrated a large language model (LLM) to generate:
 - * Attribute-specific prompts for each visual category.
 - * Context-aware descriptions to differentiate between top candidate classes.
- Achieved state-of-the-art results across 11 benchmark datasets and 5 few-shot settings.
- Applications include medical imaging and personalized AI.
- Published in *International Journal of Computational Intelligence Systems*.

RESEARCH INTERESTS

- Natural Language Processing (NLP)
- GenAI and Management
- GenAI and Cybersecurity
- Multi Modal Language Models (MMLMs)
- Prompt Engineering for Multi Modal Problems
- Explainable AI
- Agentic AI
- Agentic Multi Modal RAG

GRANTS AND FUNDING

- **Grant proposal development**, Microsoft Generative AI For Accessibility Grant , Canada, 2024
- **University Grant** for the "Design and Implementation of a 3D Printer for Printing with Bullets" – Contributed to Grant Writing and Implementation, Turkey 2022-2022
- **Co Investigator**, National Iranian Gas Company (NIGC) research Grant, "Efficient Machine Learning for Early Failure Detection" - Contributed to grant proposal development and led key aspects of project execution, 2020-2021.
- **Principal Investigator**, University Grant, "University Document Management System", 2020-2021.

PROFESSIONAL SERVICE

Editorial Roles:

- **Reviewer**, Expert Systems with Applications, International Journal of Data Science and Analytics, Engineering Applications of Artificial Intelligence, Information Processing and Management, IEEE Transactions on Multimedia, IEEE Transactions on Computational Biology and Bioinformatics, Computational Intelligence, Artificial Intelligence Review, 2022 - Present
- **Editorial board member**, of International Science Journal, 2022 -present.

Conference Reviewer for :

- IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2025
- British Machine Vision Conference (BMVC), 2024 and 2025.

Outreach and Engagement:

- **Mentored** over 30 master's students and two PhD Candidate in their research projects and thesis work, providing guidance on topic selection, methodology, and publication strategies, 2017-present.
- **Speaker**, Summit of Generative AI, Istanbul Gelisim University, 2023
- **Organizer**, International Workshop on ACCUMULATE project, Leuven, Belgium, 2020
- **Organizer**, Second Provincial Programming Hackathon , Hormozgan, Iran, 2021

CERTIFICATIONS

- Generative AI Engineering, *Certified By IBM* 2025
- University Teaching, *Carleton University* 2024
- AI for Grant Writing, *Kennesaw State University* 2025
- Explainable AI with Python, *Udemy* 2024

REFERENCES

- Associate Professor. Dr. Majid Komeili, Carleton University, majid.komeili@carleton.ca
- Professor. Dr. Hala Own, Algonquin College, hown@algonquincollege.com
- Professor. Dr. Jan Snajder, University of Zagreb, jan.snajder@fer.hr
- Professor. Dr. Ekrem Varoglu, Eastern Mediterranean University, ekrem.varoglu@emu.edu.tr
- Asst. Professor Dr. Ruhsan Onder, Cyprus Science University (now named as Cyprus Aydin University) , ruhsanonder@cau.edu.tr